

Selective Exposure, Heterogeneous Responses, and Sign Reversal in the Political Consequences of Violence*

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Abstract

Violence is unequally borne, and reactions to it diverge. Yet research on its participatory consequences rarely asks what it does to those it harms. I show that violence risk can covary with its political response, so effects can reverse sign across populations. I formalize this in a principal-stratification framework and develop a procedure that partially identifies population-wide effects and effects among victims from cross-sectional data. I apply it to reassess claims that criminal victimization increases non-electoral participation, using 97 survey waves from 20 Latin American countries. Population-wide bounds are mostly positive, consistent with prior research, whereas bounds among victims are mostly negative. Consistent with such covariance, a Mexican panel shows future victims were more exposed than neighbors before victimization and withdrew afterward. Civil war and policing reassessments show sign reversals may generalize beyond crime. When harms are selectively borne, and responses diverge, the first-order causal contrast is among those harmed.

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What would happen, on average, if everyone in the population received a treatment? This population-wide causal contrast is often taken as the benchmark for causal knowledge in empirical social science ([Winship and Morgan, 1999](#); [Angrist, 2004](#)). Indeed, the answer to this question has practical and theoretical importance regarding the effects of treatments that could, in principle, reach everyone. What would happen to mortality if a vaccine became available, or to turnout if voting became mandatory? But many treatments of interest to political science are not of this sort. Instead, they are harms that befall some and spare others.

For harmful, selective treatments, the population-wide causal contrast asks us to imagine a counterfactual of universal assignment that is unlikely to happen and that no one would wish to realize. I argue that the relevant counterfactual in such cases is instead what the harm does to those it burdens, relative to a world in which they were spared it. In causal-inference terms, the choice of counterfactual question corresponds to the choice of estimand. The population-wide contrast corresponds to the average treatment effect (ATE), while the contrast among the treated corresponds to the average treatment effect on the treated (ATT).

Violence is such a harm, and its consequences for participation are a central concern of policymakers and researchers alike. Existing theoretical accounts of those consequences point to potentially divergent reactions to it. Victimization may encourage participation in some by triggering psychological growth, engendering political grievances, or stimulating demands for more government protection ([Tedeschi and Calhoun, 2004](#); [Blattman, 2009](#); [Bateson, 2012](#)). It may demobilize others by producing fear, raising the psychological costs of public engagement, or eroding trust in politics ([Ley, 2018](#); [Trelles and Carreras, 2012](#)). Which aggregate effect we observe will depend on what violence does to those it reaches.

Extant research on violence and participation has long recognized that victims and non-victims differ across characteristics that also shape their baseline propensity to participate. Race ([Anoll, 2022](#)), age ([Plutzer, 2002](#)), socioeconomic status ([Brady, Verba and Schlozman, 1995](#)), extraversion ([Gerber et al., 2011](#)), and risk tolerance ([Kam, 2012](#)), for instance, predict distinct participatory profiles. Many of these same traits also predict exposure to violence, whether criminal victimization ([Cawvey et al., 2018](#)), police contact ([Knox, Lowe and Mummolo, 2020](#)), or armed conflict ([Balcells and Stanton, 2021](#)). Empirical studies of victimization and participation, therefore, treat such traits as confounders, since victims and non-victims may differ in their baseline participation (e.g., [Bateson, 2012](#)).

I argue that victims and non-victims may also differ in what violence would do to them, were they to

experience it. Exposure to violence follows in part from choices motivated by personal orientations and preferences, made under social and economic constraints. These same features that sort people into more or less risk of exposure to violence may also covary with how people would react to victimization, whether because these features co-produce those participatory reactions as well or because they are epiphenomena of dispositions and constraints that do. If the risk of violence covaries with the participatory reaction it would provoke, the question for empirical research is not simply how to adjust comparisons between victims and non-victims to ensure identification of an estimand. It is the choice of causal contrast to target in the first place, or equivalently, whose reactions to violence should be included.

Methodological research across disciplines has variously termed this dependence between treatment risk and potential response essential heterogeneity (Heckman, Urzua and Vytlačil, 2006), selection on gains (Roy, 1951; Heckman, 1997), and composition bias (Xie, 2013). When treatment risk covaries with potential response, estimands defined over different populations need not agree in magnitude or even in sign (Winship and Morgan, 1999; Heckman and Vytlačil, 2005). Although this result is well established in methodological work, it has rarely shaped empirical research on violence and participation. Findings interpreted as evidence of what violence does may therefore average over a different mix of potential responses than the one found among those who actually experience it.

I develop a principal-stratification framework for any binary treatment–outcome pair to formalize why effects defined over different populations can diverge when treatment risk covaries with potential response (Frangakis and Rubin, 2002; Page et al., 2015). I classify individuals by their potential outcomes under treatment and control. Unlike familiar instrumental-variables applications, where principal strata are defined by potential treatment receipt because the inferential problem is compliance with assignment (Angrist, Imbens and Rubin, 1996), my strata are defined by potential response, as in related applications in epidemiology (Greenland and Robins, 1986). This classification yields four latent response types. Two are treatment-insensitive: the outcome occurs whether treated or not, or in neither state. Two are treatment-sensitive: the outcome occurs only if treated or only if untreated. In the violence-participation application, these types correspond to always-participants, never-participants, mobilizers, and demobilizers.

Using this framework, I show that the ATE, the population-wide benchmark estimand in empirical research, depends only on the population shares of the treatment-sensitive response types. In the application,

these are the mobilizers and demobilizers. The ATT, however, weights those shares in proportion to their respective treatment probabilities. The two estimands are therefore differently weighted averages of the same underlying response structure. I further show that the ATE and the ATT coincide when treatment effects are homogeneous or when treatment probabilities are uniform across response types, as under random assignment. These are familiar equivalence conditions from the literature on heterogeneous effects and essential heterogeneity ([Winship and Morgan, 1999](#); [Heckman and Vytlačil, 2005](#)). When treatment probabilities instead differ across response types, the estimands generally diverge in magnitude and may diverge in sign.

I then derive the sign-reversal threshold in closed form. The threshold gives the degree to which the less prevalent response type must be overrepresented among the treated for the ATE and the ATT to have opposite signs. [Table 2](#) reports this threshold across ATE magnitudes and ratios of treatment probabilities. The table provides a data-agnostic, general diagnostic tool for applied researchers working with binary treatment and outcomes. Given an ATE, it shows what assumptions about response-type prevalence and differential exposure would make the effect among the treated point in the opposite direction.

I next turn to an empirical reassessment of the well-established finding that criminal victimization increases non-electoral political participation. I focus on Latin America, the most violent region in the world ([Vilalta, 2020](#)) and the setting for much of the extant empirical evidence.¹ Beginning with [Bateson \(2012\)](#), subsequent studies have largely corroborated the finding that crime spurs non-electoral political engagement ([Dorff, 2017](#); [Nussio, 2019](#); [Ponce, Somuano and Velázquez López Velarde, 2022](#)). I take at face value the strongest reading given to these findings, that victimization has a positive average effect across the target populations represented in the literature. The reassessment asks whether that conclusion survives when the effect is defined among those whom crime actually reaches.

To do so, I assemble 97 waves of the LAPOP surveys, with approximately 150,000 respondents across 20 Latin American countries ([LAPOP, 2022](#)), an extended version of the data underlying the Latin America results in [Bateson \(2012\)](#) and much of the empirical literature on the topic. These repeated cross-sections allow me to estimate the prevalence of victimization, of participation, and how participation rates differ between victims and non-victims. They do not, however, allow me to identify the quantities in my framework that determine the ATE and the ATT, including how common mobilizers and demobilizers are in the

¹A systematic review, reported in [Appendix C](#), identifies nine peer-reviewed papers that estimate the effect of individual criminal victimization on political participation. Seven draw exclusively on Latin American samples.

population and how likely each response type is to be victimized.

Instead of point-identifying the estimands, I use the framework to develop a partial-identification procedure that simultaneously bounds the population-wide effect (ATE) and the effect among victims (ATT) (Manski, 1990; Duarte et al., 2024). Recall that the framework expresses the ATE and the ATT as functions of response-type shares and type-specific victimization risks, neither of which is observable from data. What is observed in the data is the prevalence of victimization, the prevalence of participation, and participation rates among victims and non-victims. The procedure starts from these observed rates and asks which configurations of response-type shares and victimization risks could have produced them, recording the ATE and ATT implied by each. The collection of implied ATE–ATT pairs is the joint identified set.

Applied to the LAPOP data, the procedure recovers millions of admissible configurations, or ATE–ATT pairs, for each participatory outcome. Across the four measures of non-electoral participation, most configurations consistent with the data have negative ATT, ranging from 53% to 91% of admissible configurations depending on the measure. The ATE, by contrast, is positive in 63%–85% of those same configurations, in line with the mobilization finding in the literature. The one electoral-intention item, willingness to vote in the next election, follows the opposite pattern: the ATE is negative in 89% of configurations, and the ATT is positive in 95%, underscoring that response types are outcome-specific.

The partial-identification procedure returns all ATEs and ATTs compatible with the participation and victimization rates observed in the LAPOP data, but it cannot, on its own, determine which admissible configuration of response types and victimization risks actually generated the data. I therefore turn to a quarterly rotating panel of approximately 174,000 Mexican city residents observed over 27 quarters (INEGI, 2024). The panel allows me to separate exposure to crime before victimization from response after it, which cross-sections cannot. I find that soon-to-be victims are more mobile, witness more criminal and gang activity, and report more violent conflicts than their neighbors, even before becoming victims. Further, they become less mobile, less trusting of law enforcement, and more fearful after crime.

Taken together, the empirical evidence suggests that crime risk falls disproportionately on people who would participate less after victimization. However, the ATT could still be positive because it also depends on response-type shares. I therefore ask how many more mobilizers than demobilizers can exist in the population before a negative ATT is no longer sign-identified as negative. The bar is stringent because sign

identification requires that every admissible configuration in the restricted set imply a negative ATT. Protest attendance and political meeting attendance meet this bar whenever mobilizers outnumber demobilizers by no more than 4 percent. Community meetings and community problem-solving do not meet full sign identification, but the ATT is negative in 99% and 94% of admissible configurations, respectively.

Last, I ask whether estimand reversals arise in other prominent studies of violence and political mobilization. I apply the partial-identification procedure to data from [Blattman \(2009\)](#) on forced abduction by the Lord's Resistance Army in northern Uganda and from [Walker \(2020\)](#) on direct police contact in the United States. For each application, I recover the joint identified sets of ATEs and ATTs for electoral and non-electoral participation. In the police-contact data, population-wide effects on non-electoral participation are mostly positive, consistent with the reported mobilization effect. Conversely, effects among those contacted mostly point to withdrawal. For electoral participation, population-wide effects are instead mostly negative, while effects among those contacted are mostly positive. In the Uganda data, both estimands lean positive for voting, consistent with the central finding that abduction increased electoral participation. For non-electoral participation, both instead lean negative, most strongly the effects among the abducted, in contrast with the positive community-leadership effect the study reports.

The paper contributes to scholarship on estimand choice by showing why it is central to research on harms as treatments, especially when theory allows for divergent responses to such treatments. Recent work argues that estimands must be chosen to match the substantive question ([Lundberg, Johnson and Stewart, 2021](#); [Egami and Hartman, 2023](#)). Methodological work shows why that choice can change empirical conclusions, since heterogeneous effects are aggregated under different weights depending on the estimand ([Heckman and Vytlačil, 2001, 2005](#); [Słoczyński, 2022](#)). I bring these insights to harms that fall selectively and may produce divergent responses. In such settings, exposure risk may covary with the direction of response itself. Moving the causal contrast from the population to the treated can therefore change not only the group being described, but the magnitude and even the sign of the recovered effect.

Second, the paper extends insights from work on essential heterogeneity and selection on gains to violence, a treatment central to political science research. Canonical work on selection on gains studies chosen treatments, such as schooling, training, and program take-up, where units select treatment partly on the returns they expect ([Roy, 1951](#); [Willis and Rosen, 1979](#); [Björklund and Moffitt, 1987](#); [Carneiro, Hansen](#)

and Heckman, 2003). The political consequences of violence make a setting where the same dependence is likely to arise, even though no one selects to become a victim. I formalize this structure in a principal-stratification framework that classifies units by potential response and allows treatment probabilities to vary across response types. The framework yields two applied tools. The first is a closed-form sign-reversal condition, reported as a lookup table, that shows when the ATE and the ATT disagree in sign. The second is a partial-identification procedure that recovers the joint identified set of ATEs and ATTs from the moments available in any binary treatment-outcome cross-section, linking the framework to recent work on partial identification (Knox, Lowe and Mummolo, 2020; Duarte et al., 2024).

Last, the paper contributes to the literature on the participatory consequences of crime. Research on high crime contexts often finds participation depressed (Ley, 2018; Trelles and Carreras, 2012). Yet studies of individual victimization find, almost without exception, that crime victims report greater political engagement, especially in non-electoral forms of participation (Bateson, 2012; Dorff, 2017; Nussio, 2019; Ponce, Somuano and Velázquez López Velarde, 2022). The results here offer an explanation for this puzzling pattern. Read at face value, the conclusions of most individual-level studies describe causal effects defined over ad hoc populations delimited by sampling frames and estimation weights. I show that aggregate participation depends instead on what victimization does to victims. If it demobilizes those harmed, victimization may lower aggregate participation even when effects defined for other populations suggest mobilization.

1 A Principal Strata Framework

This section develops a principal-stratification framework (Frangakis and Rubin, 2002) for binary treatments whose risk may covary with potential response. The framework classifies units by their binary potential outcomes under treatment and control (Greenland and Robins, 1986), admitting individual effects of either sign, then allows each response type its own probability of treatment. I use the framework to derive expressions that link the ATE and the ATT to the shares of the response types and their risks of treatment.²

Consider a population of units indexed by i and a binary treatment $Z \in \{0, 1\}$, for example, experiencing violence or not. Denote the potential outcome for unit i under treatment assignment z as $Y_i(z)$. I classify

²Research on essential heterogeneity imposes structure on this dependence through a choice equation, or uses a plausibly exogenous instrument, to point-identify effects (Heckman and Vytlačil, 2005; Heckman, Urzua and Vytlačil, 2006; Brand and Xie, 2010; Zhou and Xie, 2019). I depart from this tradition, as the end goal here is not point identification but partial identification. This dispenses with the assumptions point identification requires, monotonicity of response, an exclusion restriction, or a model of selection into treatment, at the price of recovering a set rather than a point.

units into strata (θ) defined by their joint potential outcomes $(Y_i(1), Y_i(0))$ (Greenland and Robins, 1986). By construction, all units in a stratum share the same individual treatment effect, $ITE_i = Y_i(1) - Y_i(0)$.

Denote the proportion of units in each stratum π_θ , with $\sum_\theta \pi_\theta = 1$. I further allow units to have stratum-specific probabilities of receiving treatment, ρ_θ . Variation in ρ_θ across strata encodes the dependence between treatment risk and potential response that choice models in the selection on gains literature generate through sorting on expected returns (Heckman, Urzua and Vytlačil, 2006).

Table 1 synthesizes the setup for the case of a binary outcome $Y_i(z) \in \{0, 1\}$, such as whether an individual participates in a given political activity or not.

Stratum (θ)	Share	Pr(Z=1)	Y(1)	Y(0)	ITE
Always Participate	π_A	ρ_A	1	1	0
Participate if Treated	π_T	ρ_T	1	0	1
Participate if Untreated	π_U	ρ_U	0	1	-1
Never Participate	π_N	ρ_N	0	0	0

Table 1: Stratum-specific individual treatment effects and treatment probabilities. Each row defines a principal stratum by its potential outcomes under treatment and control. The treatment probability ρ_θ is allowed to vary across strata.

For a binary treatment such as exposure to violence, we can classify individuals into four strata: those who always participate (A), those who never participate (N), those who participate if and only if treated (T), and those who participate if and only if untreated (U).³

Proposition 1. *The average treatment effect is $ATE = \pi_T - \pi_U$.*

Proposition 1 states that the ATE depends only on the relative shares of strata T and U , an identity that follows directly from the response-type partition (Greenland and Robins, 1986). This result is straightforward, as all units in strata A and N contribute an ITE of zero. Consequently, the magnitude and direction of the ATE will only depend on the frequency with which T appears in the population relative to U .⁴

³I make the stable unit treatment value assumption (SUTVA) throughout; the framework models responses to one's own victimization, not spillovers from others'.

⁴Formal proofs for all expressions in this section are provided in Appendix A.

Proposition 2. *The average treatment effect on the treated is*

$$ATT = \frac{\pi_T \rho_T - \pi_U \rho_U}{Pr(Z = 1)},$$

where $Pr(Z = 1) = \sum_{\theta} \pi_{\theta} \rho_{\theta}$.

Conversely, the ATT, or the average expected change in outcomes for those who are treated, will be a function of population shares and also of the stratum-specific probabilities of receiving treatment. These stratum-specific probabilities scale the relative contributions of the T and U strata within the treated population. Intuitively, strata with higher treatment probabilities constitute a larger share of the treated group and therefore should contribute more to the ATT. Further rearranging the numerator of the expression for the ATT yields an identity for the population-level impact of treatment:

$$ATT \times Pr(Z = 1) = \pi_T \rho_T - \pi_U \rho_U. \quad (1)$$

I denote this quantity ΔY , the average change in Y attributable to the realized treatment assignment.

Corollary 1. *The ATE and ATT are equal when $\rho_T = \rho_U = Pr(Z = 1)$.*

Intuitively, when $\rho_T = \rho_U = Pr(Z = 1)$, the responsive strata are treated at the population rate, so conditioning on being treated does not alter the relative representation of any stratum. That is, the treated group is a random draw from the population, so the ATE and the ATT average over identical weights in expectation.⁵

Corollary 2. *When $\rho_T = \rho_U$, the ATE and ATT have the same sign. Specifically, $ATT = \frac{\rho_T}{Pr(Z=1)} \cdot ATE$.*

Corollary 2 relaxes Corollary 1 by allowing the treatment-insensitive strata (A and N) to have treatment probabilities that differ from those of the treatment-sensitive strata. Because the A and N strata have a treatment effect of zero, their over- or under-representation in the treated group cannot change the sign of the ATT relative to the ATE. Only the relative weighting of T and U determines the sign, so $\rho_T = \rho_U$

⁵This condition, derived here from the response-type partition alone, is equivalent to extant results from research on heterogeneous treatment effects and essential heterogeneity. Heckman and Vytlačil (2005: p. 681), Heckman and Vytlačil (2001: pp. 107, 109), Angrist (2004: pp. C63–C64), and Winship and Morgan (1999: pp. 666–667) each show that estimands coincide when effects do not vary across units or when treatment is unrelated to how they vary.

suffices for sign agreement. The treatment-insensitive strata do, however, affect the *magnitude*: the scaling factor $\rho_T / Pr(Z = 1)$ can amplify or attenuate the ATT but cannot reverse its direction. Corollary 1 is the special case where $\rho_T = Pr(Z = 1)$, so the scaling factor equals one.

When do the ATE and ATT disagree in sign? The ATE and ATT need not agree in sign. Signs agree when composition and selection reinforce each other: for example, when the T -stratum is both larger than the U -stratum and at least as likely to be treated, or when the U -stratum is both larger and at least as likely to be treated. Signs can diverge when one stratum is larger but the other is more likely to be treated, so that the stratum that dominates the population average is under-represented among the treated. I now characterize exactly when this occurs.

To do so, I define two summary parameters. Let $\lambda = \rho_U / \rho_T$ capture differential selection between the responsive strata. When $\lambda > 1$, the U -stratum is more likely to be treated than the T -stratum; when $\lambda < 1$, the reverse holds. Last, when $\lambda = 1$, we have equal selection and necessarily sign agreement (Corollary 2). Let $\eta = \pi_T / \pi_U$ capture the relative size of the two responsive strata. Hence $\eta > 1$ when the T -stratum is larger ($ATE > 0$) and $\eta < 1$ when the U -stratum is larger ($ATE < 0$).

Proposition 3. *Let $\pi_T > 0$ and $\pi_U > 0$. Then $\text{sign}(ATT) = \text{sign}(\pi_T \rho_T - \pi_U \rho_U)$, and the ATE and ATT have opposite signs if and only if $\lambda > \eta$ when $ATE > 0$, or $\lambda < \eta$ when $ATE < 0$.*

The sign of the ATT is determined by whether the T -stratum's exposure-weighted size ($\pi_T \rho_T$) exceeds that of the U -stratum ($\pi_U \rho_U$). When $\lambda > 1$, the U -stratum is over-represented among the treated, pulling the ATT toward negative values. A sign reversal occurs when this differential selection more than compensates for the size advantage of the T -stratum, that is, when $\lambda > \eta$. The condition is symmetric: when $ATE < 0$ ($\eta < 1$), the ATT is positive only if $\lambda < \eta$. In that case, the T -stratum must be sufficiently more exposed to treatment than the U -stratum to overcome its smaller population share.

Corollary 3. *When $ATE > 0$ and $\lambda > 1$:*

$$ATT < 0 \iff \pi_U > \frac{ATE}{\lambda - 1}. \quad (2)$$

By symmetry, when $ATE < 0$ and $\lambda < 1$, the ATT is positive if and only if $\pi_T > |ATE| \cdot \lambda / (1 - \lambda)$.⁶

⁶Define $\lambda' = \rho_T / \rho_U = 1/\lambda$ and $\eta' = \pi_U / \pi_T = 1/\eta$. The sign-flip condition $\lambda < \eta$ is equivalent to $\lambda' > \eta'$, and the same

For the sign of the ATT to oppose the sign of the ATE, the countervailing stratum must constitute a sufficiently large share of the total population. How large depends on (i) the magnitude of the ATE, and (ii) how much more exposed the countervailing stratum is to treatment. When the ATE is small or λ is far from one, the threshold is low, so sign reversal requires only a modest population share of the countervailing stratum. Table 2 illustrates this relationship for a range of ATE magnitudes and selection ratios λ .

Consider the bottom-left region of the table, where the ATE is positive and the T -stratum is also more likely to be treated ($\lambda < 1$). These cells show check marks when composition and selection reinforce each other, so the ATE and ATT always agree in sign regardless of stratum shares. Now consider the bottom-right region, where the ATE is still positive but the U -stratum is more likely to be treated ($\lambda > 1$). Here, the threshold cells show the minimum population share of the U -stratum (π_U , as a fraction of the total population) above which the ATT turns negative. For example, at $ATE = 0.04$ and $\lambda = 2$ (meaning each member of the U -stratum is twice as likely to be treated as each member of the T -stratum), the ATT is negative whenever the U -stratum constitutes more than 4% of the population. Conversely, the top-left region applies when the ATE is negative and the T -stratum is more likely to be treated ($\lambda < 1$). At $ATE = -0.10$ and $\lambda = 1/3$ (meaning each member of the T -stratum is three times as likely to be treated as each member of the U -stratum), the ATT is positive whenever the T -stratum exceeds 5% of the population.

algebra yields $\pi_T > |ATE|/(\lambda' - 1) = |ATE| \cdot \lambda/(1 - \lambda)$.

Selection ratio $\lambda = \rho_U / \rho_T$							
	<i>T</i> more likely treated			1	<i>U</i> more likely treated		
	$\frac{1}{3}$	$\frac{1}{2}$	$\frac{2}{3}$		$\frac{3}{2}$	2	3
<i>ATE</i>	−0.10	$\pi_T > 5\%$	$\pi_T > 10\%$	$\pi_T > 20\%$	✓	✓	✓
	−0.04	$\pi_T > 2\%$	$\pi_T > 4\%$	$\pi_T > 8\%$	✓	✓	✓
	−0.02	$\pi_T > 1\%$	$\pi_T > 2\%$	$\pi_T > 4\%$	✓	✓	✓
	0.02	✓	✓	✓	✓	$\pi_U > 4\%$	$\pi_U > 2\%$
	0.04	✓	✓	✓	✓	$\pi_U > 8\%$	$\pi_U > 4\%$
	0.10	✓	✓	✓	✓	$\pi_U > 20\%$	$\pi_U > 10\%$

Table 2: Minimum share of the countervailing stratum required for the ATT to have opposite sign from the ATE, as a function of the ATE and the selection ratio $\lambda = \rho_U / \rho_T$. When $\lambda > 1$, the *U*-stratum is more likely to be treated than the *T*-stratum; when $\lambda < 1$, the reverse. Cells show the threshold from Corollary 3. Off-diagonal check marks indicate regions where differential selection reinforces population share differences, so the ATE and ATT always agree in sign. The $\lambda = 1$ column indicates no differential selection, so signs agree by Corollary 2.

The framework and Table 2 are entirely data-agnostic, as they apply to any setting with a binary treatment and a binary outcome in which responses can run in either direction and treatment risk covaries with potential response. Any empirical researcher can use the table to assess, for a given ATE magnitude and belief about differential selection, whether the ATE and ATT might diverge in sign. In the next section, I apply this framework to criminal victimization and political participation.

2 Application to Criminal Victimization

Extant theoretical accounts of the participatory consequences of criminal victimization, drawing heavily from work on civil war and policing violence, posit that victimization may affect participation through two competing channels. The first encompasses processes that increase participation. Post-traumatic growth may transform victims into civic actors, anger may generate demands for political action, shared victimization may build solidarity and lower collective-action costs, and victims may acquire instrumental incentives to demand security or punish incumbents (Bateson, 2012; Blattman, 2009; Koos and Traummüller, 2024;

Gilligan, Pasquale and Samii, 2014; Ley, 2022). The second encompasses processes that suppress participation. By engendering fear, crime may raise the perceived costs of public engagement. Victimization may also erode trust in the capacity of the state to provide protection (Ley, 2018; Hale, 1996). These two broad channels, and the many mechanisms they encompass, map onto the *Participate if Treated* and *Participate if Untreated* strata of Table 1, respectively.

Despite these competing theoretical predictions, the empirical literature is remarkably uniform in its conclusion that criminal victimization increases non-electoral civic and political engagement (Bateson, 2012; Dorff, 2017; Ley, 2022). I conducted a systematic review of peer-reviewed studies that estimate the individual-level relationship between criminal victimization and some form of political participation. The review identified nine papers, spanning twenty-one distinct non-electoral outcomes. Every significant estimate on those outcomes is positive. Two papers additionally draw on within-person variation and find significant positive effects there as well, one of them for electoral turnout (Laterzo, 2021; Sønderskov et al., 2022). The evidentiary basis for this consensus is also geographically concentrated. Seven of the nine papers draw exclusively on Latin American samples, and four analyze waves of the LAPOP surveys (LAPOP, 2022), an extended version of the data I examine in the next section.⁷

All of these studies recognize that victimization risk is not randomly distributed (Cohen and Felson, 1979; Gottfredson, 1986). Risk varies systematically with mobility, risk preferences, socioeconomic status, and access to protection (Miethe, Stafford and Long, 1987). Wealthier and more risk-averse individuals tend to be better able or more willing to avoid victimization (Skogan, 1995; Hale, 1996), whereas more mobile and risk-prone individuals embedded in their communities face greater risk and often weaker protection (Rader, 2004; Browning, Pinchak and Calder, 2021; Boggs, 1965; Guedes, Domingos and Cardoso, 2018). These same characteristics also predict political participation (Gerber et al., 2011; Kam, 2012). Indeed, the same profiles predict both victimization and participation in Latin America (Cawvey et al., 2018).

Existing research has treated the joint determinants of victimization and participation chiefly as a problem of identification. The usual concern is that victims and non-victims may differ in how much they would have participated had neither group been victimized. Researchers, therefore, adjust for observed characteristics associated with both victimization risk and participation. The causal interpretation of the cross-sectional

⁷See Appendix C for the search strategy, inclusion criteria, and coding decisions.

comparisons on which much of the evidence rests depends on the assumption that this adjustment removes the relevant differences between victims and non-victims.

I argue that existing research has overlooked a second dimension of selection. Victims and non-victims may differ not only in how much they would participate without victimization, but also in how victimization would change their participation. Criminal victimization may therefore involve selection on responses. The same dispositions and constraints that shape victimization risk may also shape the response victimization produces. While this second dimension complicates identification because adjustment must account for differences in potential responses and in untreated outcomes,⁸ the more fundamental implication of selection on responses concerns the causal effect researchers should target.

If treatment risk covaries with potential response, a population-wide effect and an effect among victims average different balances of positive and negative responses (Propositions 1 and 2) and therefore need not coincide in sign or magnitude (Proposition 3). The question is then not only whether victim–non-victim comparisons identify a causal effect, but whether the effect of interest describes the population at large or the people crime actually reaches. That is, what causal effect to aim at to begin with.

The analyses in the next section isolate the consequences of that estimand choice. I take the prevailing population-wide interpretation at face value and ask whether the conclusion that crime mobilizes survives when the causal effect is instead defined among victims.

2.1 Data

I focus on Latin America, where victimization is widespread (Vilalta, 2020; LAPOP, 2022) and where much of the research on the political consequences of crime has been conducted (Visconti, 2020; Ley, 2018; Blanco, 2013; Marshall, 2022; Kronick, 2014). As the primary data source, I employ an extended version of the LAPOP survey, which Bateson (2012) uses for their Latin American results. I pool all surveys conducted in 20 Latin American countries between 2010 and 2019, yielding approximately 150,000 respondents from 97 survey waves. Each survey round was designed to represent the voting-age population of the country in that year.⁹ LAPOP measures self-reported victimization in the past 12 months and several forms of civic and political engagement. I focus on five outcomes: protest participation, political meeting attendance,

⁸Section A3.1 examines these assumptions for the designs reviewed here in light of the framework in Section 1.

⁹See A2.1.1 for sampling details and A2.4 for human-subjects compliance.

community meeting attendance, participation in community problem-solving activities, and willingness to vote in upcoming national elections if they were held next week.

2.2 Empirical Strategy

I implement a partial identification procedure using the LAPOP data to characterize the set of ATEs and ATTs consistent with four observed moments of the joint distribution of victimization and participation. From the pooled data, I can retrieve the proportion of respondents who report being victims, $Pr[Z = 1]$ and its converse probability, the proportion of respondents who report engaging in each of the participatory activities, $Pr[Y = 1]$, and its converse, and the proportion of victimized/unvictimized respondents who report participating $Pr[Y = 1|Z = 1]$, $Pr[Y = 1|Z = 0]$, respectively. Table 3 reports the observed probabilities in the data for each of the outcomes.¹⁰

Leveraging the structure of the principal-strata framework, these observed quantities restrict the possible values of the population shares of each stratum (π_θ) and their corresponding treatment probabilities (ρ_θ). The expressions derived in Section 1 imply that any feasible parameter vector must reproduce the observed participation moments through the identities linking $(\pi_\theta, \rho_\theta)$ to $(Pr[Y = 1], Pr[Y = 1 | Z = 1], Pr[Y = 1 | Z = 0], Pr[Z = 1])$. For each feasible vector, I then compute the ATE and ATT directly using the closed-form expressions established earlier:

$$ATE = \pi_T - \pi_U, \quad ATT = \frac{\pi_T \rho_T - \pi_U \rho_U}{Pr(Z = 1)}.$$

To characterize the full set of parameter values consistent with the data, I explore the feasible parameter space for $\{\pi_\theta, \rho_\theta\}$ for all $\theta \in \{A, T, U, N\}$. The approach allows me to recover the entire identified set of parameter combinations implied jointly by the observed moments and the model's structure (Kline and Tamer, 2023). This proceeds as follows.

1. I construct a fine grid over all possible population shares $(\pi_A, \pi_T, \pi_U, \pi_N)$ so $\sum_\theta \pi_\theta = 1$.
2. For each candidate vector of population shares, I use the observed moments in Table 3 to solve for the implied treatment probabilities ρ_T and ρ_U that rationalize the observed conditional participation

¹⁰These proportions are estimated from survey data and therefore contain uncertainty. However, the large sample size yields very precise estimates, rendering the incorporation of this uncertainty negligible and immaterial to the results. For simplicity and clarity, I omit this uncertainty from the explanation.

Participation Type	N	$Pr(Z = 1)$	$Pr(Y = 1)$	$Pr(Y = 1 Z = 1)$	$Pr(Y = 1 Z = 0)$
Attended Protest	149,728	0.21 (0.001)	0.08 (0.0007)	0.14 (0.002)	0.07 (0.0007)
Community Problem-Solving	94,022	0.19 (0.001)	0.36 (0.001)	0.43 (0.004)	0.34 (0.002)
Attended Community Meeting	149,405	0.21 (0.001)	0.29 (0.001)	0.32 (0.003)	0.28 (0.001)
Attended Political Meeting	137,308	0.20 (0.001)	0.16 (0.0009)	0.18 (0.002)	0.15 (0.001)
Would vote next week	127,256	0.22 (0.001)	0.81 (0.001)	0.85 (0.002)	0.80 (0.001)

Table 3: Descriptive statistics from the LAPOP survey for each measure of political participation. N : respondents with non-missing participation and victimization responses. $Pr(Z = 1)$: victimization rate. $Pr(Y = 1)$: unconditional participation rate. $Pr(Y = 1 | Z = 1)$ and $Pr(Y = 1 | Z = 0)$: participation rates among victims and non-victims, respectively. Standard errors in parentheses.

rates.

3. I then solve for ρ_A and ρ_N using the marginal probability of victimization, $Pr(Z = 1)$.
4. Any parameter vector that violates logical or probabilistic constraints (e.g., treatment probabilities outside $[0, 1]$ or participation rates inconsistent with potential outcomes) is discarded.
5. For every remaining feasible parameter vector, I compute the corresponding ATE and ATT using the expressions above, as well as the implied differential selection ratio $\lambda = \rho_U / \rho_T$ and stratum-size ratio $\eta = \pi_T / \pi_U$, which jointly govern whether the ATT agrees in sign with the ATE (Section 5).

The procedure yields the identified set, that is, the set of parameter vectors consistent with the LAPOP moments. Each vector implies an ATE, ATT pair along with stratum-specific treatment probabilities, and stratum-specific population shares. Together, these vectors characterize the causal effects compatible with the data and the framework. Whenever individuals who would reduce participation after victimization are more likely to be victimized (i.e., $\rho_U > \rho_T$), the ATE and ATT diverge, often in magnitude and sometimes in sign. All derivations and computational details are reported in [Appendix D](#).

3 Results: Crime Victimization

After repeating the algorithm detailed in the previous section for each of the participatory outcomes, I obtained 12.1 million, 16 million, 20.6 million, 18.34 million, and 3.63 million feasible parameter vectors $\{\pi_\theta, \rho_\theta\} \forall \theta \in \{A, T, U, N\}$, each implying a specific λ and η , for the protest attendance, political meeting attendance, community meeting attendance, community problem-solving outcomes, and willingness to vote respectively. These are an internally consistent set of values that bound the actual value of each parameter.

The difference in the sizes of the parameter sets depends on how informative the data is; for infrequent outcomes, fewer parameter sets are consistent with the data.

For each set, I can then calculate the corresponding ATE and ATT for victimization on each measure of political participation. Figure 1 plots the results. Each point in the graph corresponds to a feasible combination of π_θ, ρ_θ for a given participatory outcome. The red diagonal line that passes through the origin indicates the points where the ATE and the ATT are equal. For all the points below, the ATE is larger than the ATT. Conversely, the ATT is larger than the ATE for all points above that line. The color of the point indicates the difference between the probability of becoming a victim for individuals who become mobilized due to victimization and the probability of becoming a victim for individuals who respond by decreasing political participation in the set.

The implied ATE is almost always larger than the corresponding ATT across all measures of non-electoral political participation. When the outcome is participating in a protest, the ATE is larger than the ATT in 99% of the parameter sets.¹¹ When attending political meetings is the outcome, the same is true for 96% of the parameter sets. When it is attending community meetings, the ATE is larger in 83% of parameter sets. When community problem-solving activities is the outcome, the percentage is 64%. Further, we can see that the points above the red diagonal line correspond to the largest differences between $\rho_T - \rho_U$. For the ATT to be larger than the ATE, selection into victimization has to be strongly correlated with the treatment, making individuals more participative.

While most configurations compatible with the data yield a positive ATE for all four measures of non-electoral participation (85% for protest attendance, 79% for political meeting attendance, 68% for community meeting attendance, and 63% for community problem solving), the converse is true for the ATTs across measures. Specifically, only 9% of compatible configurations yield a positive ATT for protest attendance, only 13% for political meeting attendance, 29% for community meeting attendance, and 47% for community problem-solving. Even for the most frequently reported outcome, community meeting attendance, most configurations compatible with the data yield a negative ATT. As reported in Table 3, the larger share of positive ATEs and negative ATTs reflects the conditional probabilities of participating in these activities.

The results from the willingness-to-vote question, if elections were held the following week, show that

¹¹Shares weight all feasible vectors uniformly, so they read as posterior probabilities only under a uniform prior. I remain agnostic and interpret them as describing the composition of the identified set.

electoral participation shows a different pattern than non-electoral forms of engagement. While the ATE exceeds the ATT across most non-electoral participation measures, this relationship reverses for voting intention. Here, the ATE exceeds the ATT in only 1% of parameter sets. More strikingly, in 89% of parameter sets compatible with the data, the ATE of victimization on willingness to vote is *negative*, while in 95% of the sets, the ATT is positive.

Three considerations help explain this reversal between electoral and non-electoral participation. First, voting may be relatively less costly than non-electoral participation. For the marginal individual, it requires no collective organization, no sustained community presence, and no resources beyond showing up. The mechanisms that suppress costly participation after victimization may therefore not suppress voting. Second, strata may be outcome-specific. An individual can be in the *Participate if Untreated* stratum for protest but in the *Participate if Treated* stratum for voting. With roughly 81% of respondents reporting willingness to vote and 22% victimization, the population is dominated by *Always* voters for whom voting behavior is unaffected by victimization, consistent with the empirical stickiness of electoral participation (Coppock and Green, 2016). Third, the reversal is consistent with extant evidence. Sønderskov et al. (2022) find that victimization increases turnout by 2 to 3 percentage points within individuals but that the between-individual association is large and negative.

The results derived in Section 1 show the divergence between the ATE and ATT could be driven by the relative size of the mobilized and demobilized strata, their differential exposure to victimization, or a combination of both. Figure 2 plots the cumulative distribution of these two quantities across all parameter vectors compatible with the LAPOP data. The left panel shows $\eta = \pi_T/\pi_U$, the ratio of mobilizers to demobilizers; the right panel shows $\lambda = \rho_U/\rho_T$, the ratio of demobilizer to mobilizer victimization risk. Both panels use a log scale, with the dashed line indicating the reference value of 1.

The left panel shows that across all four non-electoral outcomes, the vast majority of the identified set has $\eta > 1$, with mobilizers outnumbering demobilizers in the population at large. Protest has the highest η values, with a median around 3.5; the remaining non-electoral outcomes cluster lower, with medians between 1.7 and 2.6. The cumulative proportion at which each curve crosses the dashed line equals the share of vectors with $ATE > 0$, since $ATE = \pi_T - \pi_U$. For voting, the pattern reverses and $\eta < 1$ for most of the identified set, indicating that demobilizers outnumber mobilizers and that the ATE is predominantly

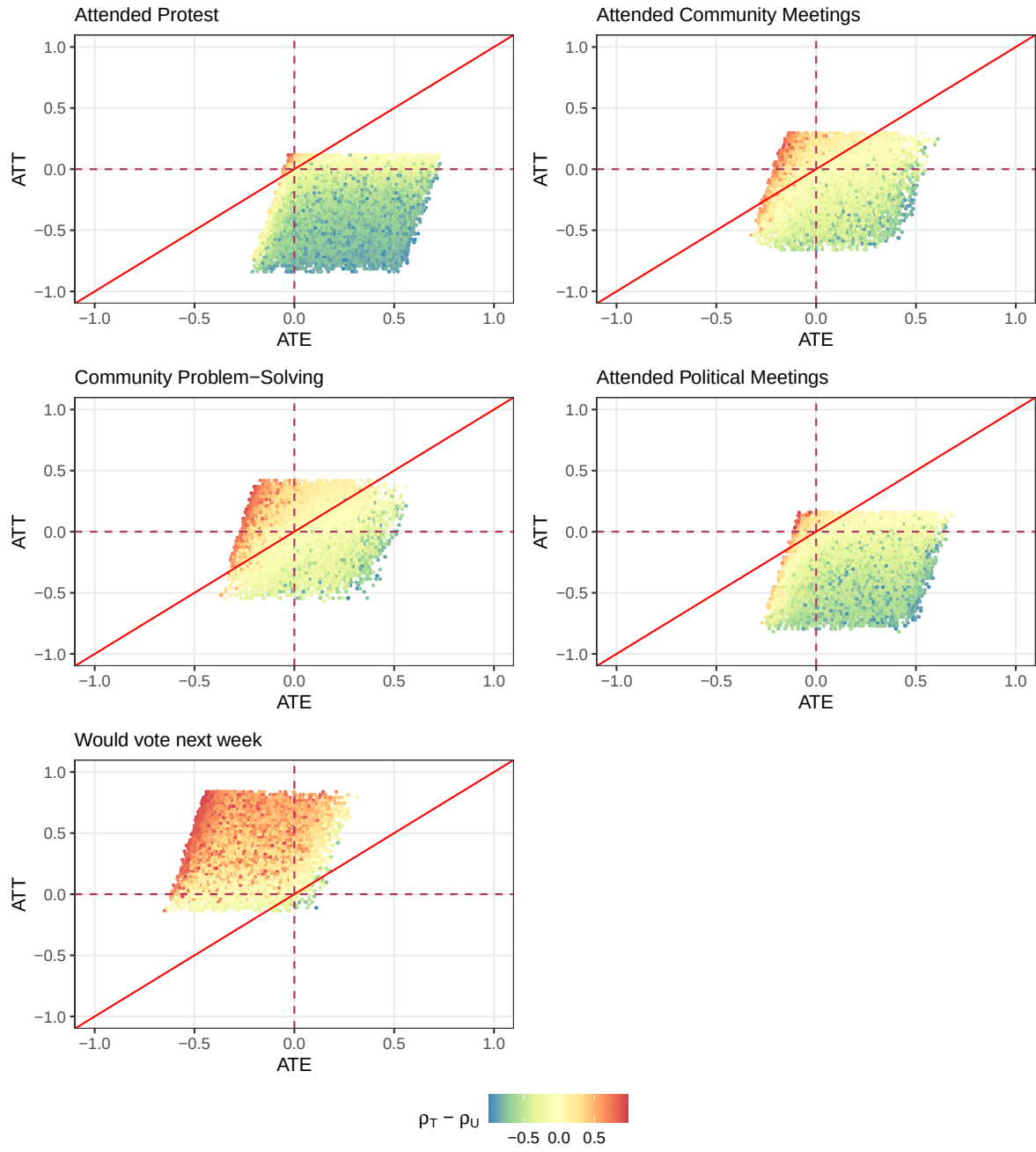


Figure 1: Figure shows the combination of potential ATTs and ATEs of victimization on five measures of political participation, conditioning on observed participation, and criminal victimization (LAPOP, 2022).

negative.

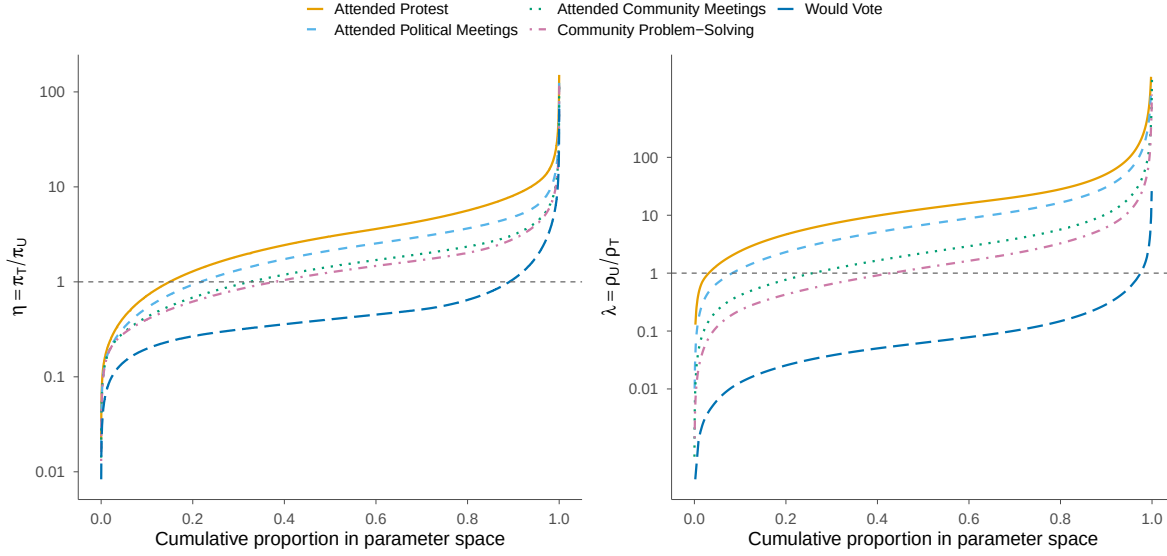


Figure 2: Cumulative distribution of the stratum-size ratio $\eta = \pi_T / \pi_U$ (left) and the differential selection ratio $\lambda = \rho_U / \rho_T$ (right) across parameter vectors compatible with the LAPOP data. Log scale. The dashed line marks 1: above it, mobilizers outnumber demobilizers (left), or demobilizers face a higher risk of victimization (right). The cumulative proportion at which a curve crosses 1 equals the share of the identified set with $ATE > 0$ (left) or $\rho_U > \rho_T$ (right).

The right panel tells the complementary story. For all four non-electoral outcomes, $\lambda > 1$ across the vast majority of the identified set, meaning demobilizers face a higher victimization risk than mobilizers. Protest and political meetings show the strongest pattern, with λ rising above 1 early in the cumulative distribution. Voting again reverses: $\lambda < 1$ for most vectors, meaning mobilizers are more likely to be victimized than demobilizers.

The results suggest that if the probability of victimization were uniform across the entire population, it would likely increase non-electoral participation, consistent with [Bateson \(2012\)](#) and the field's prevailing understanding.¹² However, because individuals who would participate less after victimization are disproportionately likely to become victims, the identified sets suggest that crime reduces all four types of non-electoral participation in the region. The pattern reverses for willingness to vote: crime, were it random, would likely reduce willingness to vote, but given actual patterns of victimization, it likely increases it.

¹²Results in [Bateson \(2012\)](#) are based on one wave of the LAPOP data. Section [A2.3](#) in the Appendix replicates the results from [Bateson \(2012\)](#) with repeated cross-sections of the LAPOP data, as well as within-country analyses.

Participation Type	$Pr(Z = 1)$	ATE			ATT			ΔY	
		Min	Max	Share $ATE < 0$	Min	Max	Share $ATT < 0$	Min	Max
Attended Protest	0.21	-0.23	0.76	0.15	-0.84	0.12	0.91	-0.18	0.02
Community Problem-Solving	0.19	-0.38	0.61	0.37	-0.57	0.42	0.53	-0.11	0.08
Attended Political Meetings	0.20	-0.28	0.70	0.21	-0.82	0.16	0.87	-0.17	0.03
Attended Community Meetings	0.21	-0.36	0.63	0.32	-0.68	0.30	0.71	-0.14	0.06
Would vote next week	0.22	-0.65	0.33	0.89	-0.13	0.84	0.05	-0.03	0.18

Table 4: Identified sets of ATE, ATT, and $\Delta Y = ATT \times Pr(Z = 1)$ from all feasible parameter vectors compatible with the LAPOP data, without constraining ρ_U or ρ_T . Min and Max report the bounds of each identified set. Share columns report the share of feasible parameter vectors for which each estimand is negative.

What are the magnitudes of the changes in participation? Recall from expression (1) that the realized change in population outcomes equals the ATT scaled by the probability of treatment. Table 4 synthesizes all the results. These unrestricted sets are the sharpest bounds the observed moments alone permit (A1.5). Participation in protests could decrease by as much as 18 pp, engagement in community problem-solving could decrease by 11 pp, political meeting attendance could decrease by 17 pp, while community meeting attendance could decrease by as much as 14 pp. Willingness to vote next week, however, at worst could decrease by 3 pp. Conversely, conditional on the data, protest attendance could *at best* increase only by 2 pp; community problem solving by 8 pp; political meeting attendance by 3 pp; and community meeting attendance by 6 pp. Hypothetical willingness to vote could increase by at most 18 pp. The *upper bound effects* within the identified set of non-electoral participation estimands are comparable in magnitude to, and often smaller than, the ATEs identified in Bateson (2012).

One concern is that these results reflect peculiarities of the LAPOP data. For example, we might be worried that LAPOP over-represents urban areas, under-represents dangerous areas, or yields survey responses that do not generalize. To address such concerns, I analyze data from Dorff (2017), who conducted a nationally representative survey of 1,000 Mexican respondents that included questions about victimization and both electoral and non-electoral participation. I apply the same partial identification procedure from the main paper to this alternative dataset. I present the results in Table A8 in the Appendix. I find strikingly similar results.

The partial identification procedure that underpins all results in this section takes the observed moments

as given. A reasonable concern is that self-reported victimization is subject to differential *reporting* that covaries with political participation, with politically engaged individuals more likely to classify an experience as criminal (Skogan, 1982; Boulding, Mullenax and Schauer, 2022). A6.1 formalizes how such bias would propagate through the framework. If it takes the form assumed in the literature, where the politically active are more likely to report, it would inflate both the ATE and the ATT, making a negative ATT harder to detect. Appendix F also presents evidence that self-reported victimization tracks municipal homicide rates in Mexico, Brazil, and Colombia, and that Mexican respondents who report being asked for a bribe are also more likely to report being extorted, suggesting that crime experience maps onto reporting, assuaging concerns that the measure is not construct-valid.

4 Panel Evidence from Mexico

The previous section established that demobilization among victims is consistent with the majority of the parameter space arising from pooled data from 20 Latin American countries. The natural next question is where in that space we actually are. Answering it requires evidence on who is selected into victimization and how they respond, which cross-sectional data cannot provide. I turn to a panel survey from Mexico, one of the countries in the LAPOP sample and the second largest country in Latin America, that follows the same individuals over time.

I use a quarterly rotating panel survey of Mexican city residents that asks questions related to crime and insecurity, the National Survey of Urban Public Safety, ENSU, for its name in Spanish (INEGI, 2024). This survey has been conducted four times per year since 2013¹³ and, starting in 2017, has included direct questions regarding personal victimization. This rolling survey panel allows me to observe individuals at five different points over 15 months. In each quarter, respondents are asked about experiences with the state, crime, and conflict, and in at least one and usually two waves per year, they are asked about direct victimization. I keep all survey waves from 2017, when crime victimization was first incorporated, through the last quarter of 2023, resulting in 560k responses from 174K unique respondents collected over 27 quarters.¹⁴

Unlike the LAPOP data, the panel structure of ENSU allows me to provide direct evidence on both

¹³Except the second quarter of 2020 due to the COVID emergency.

¹⁴See A2.1.2 for sampling details.

conditions that the framework requires. First, by comparing future victims with non-victims before any victimization occurs, I assess whether exposure is selective. If it is, future victims should differ systematically from their neighbors even before becoming victims. Second, by exploiting within-individual variation, I estimate how victimization changes attitudes and behavior among those who experience it, providing evidence on the direction of response among the treated.¹⁵

To assess whether exposure to victimization is selective, I identify ENSU questions that map onto the characteristics related to victimization risk: mobility patterns, exposure to criminal and violent behavior, feelings of unsafety, and interpersonal conflicts. Higher mobility increases exposure to crime (Cohen and Felson, 1979; Miethe, Stafford and Long, 1987; Rader, 2004; Browning, Pinchak and Calder, 2021); witnessing criminal behavior directly indexes victimization risk (Gottfredson, 1986); and fear of crime correlates with perceived vulnerability (Hale, 1996; Guedes, Domingos and Cardoso, 2018). Importantly, these same characteristics are also correlates of political participation, as extroverted, mobile, community-oriented, and risk-prone individuals participate more (Gerber et al., 2011; Kam, 2012; Ley, 2022). In Latin America, these are also the individuals who are victimized more often (Cawvey et al., 2018).

To assess reactions to victimization, I estimate how victimization changes attitudes and behavior among those who experience it, several of them the same measures that distinguished future victims beforehand. I exploit within-individual variation with two estimators, a two-way fixed effects estimator and the fixed-effect counterfactual estimator of Liu, Wang and Xu (2022), focusing on mobility, social contact, crime-prevention behavior, fear of crime, and trust in law enforcement. If victimization pushes victims toward withdrawal, they should leave home less, guard themselves more, and become more fearful. These are the changes the participation literature ties to disengagement from public life, consistent with *Participate if Untreated* reactions.

4.1 Exposure to Crime

I subset the data and retain only respondents who never reported criminal victimization and respondents who reported criminal victimization in some, but not all, survey waves. For the latter, I retain responses

¹⁵Like any survey measure, self-reported victimization may be subject to heterogeneous reporting: individuals may interpret events as criminal or non-criminal differently, and such sensitivity could correlate with political attitudes and behavior (Skogan, 1982; Boulding, Mullenax and Schauer, 2022). Two descriptive analyses in Appendix F show that self-reported victimization tracks municipal-level criminal victimization rates as expected.

only before their first reported instance of victimization. Consequently, I compare the responses of this group of future victims *before they reported victimization* with those of their never-victimized neighbors, either from the same sampling unit or the same locality, in the same quarter.

I estimate

$$PM_{i[q,g]} = \beta_1 * FutureVictim_{i[q,g]} + \theta_{q \times g} + \epsilon_{i[q,g]} \quad (3)$$

where $PM_{i[q,g]}$ is respondent i 's self-reported behavior, experiences, or attitude in year-quarter q , and neighborhood/census tract or locality g , or a binary measure that takes the value of 1 if respondent i reports that type of behavior or experience and 0 otherwise. $FutureVictim_{i[q,g]}$ is a binary variable that takes the value of 1 when respondent i will report having been a victim of crime in a future survey and 0 otherwise. $\theta_{q \times g}$ represent either *Neighborhood* $\times$ *Quarter* fixed effects, or *Locality* $\times$ *Quarter* fixed effects. $\epsilon_{i[q,g]}$ are robust errors clustered at the primary sampling unit.

Future victims differ systematically from non-victims before victimization occurs (Figure 3). When we compare future victims with never-victims living in the same locality in that same quarter (dark blue) or the same small neighborhood (light blue), future victims report having more conflicts, including conflicts that resulted in physical violence, conflicts with authorities, conflicts with the police, and conflicts with strangers on the street, *before* being victimized. Future victims also witnessed gangs, robberies, and vandalism, and heard gunshots more frequently. They also report feeling unsafe in the city, on the streets, on public transport, and even in their house. Last, they report leaving their home more frequently.

Overall, the results link future victimization with increased exposure to crime and show that victims and non-victims were differentially exposed to crime even before becoming victims. Future victims leave home more frequently and report more interpersonal conflicts, including conflicts with authorities, police, and strangers, than their neighbors in the same census tract and quarter. By comparing neighbors to each other, we can ensure that the results are not mechanically driven by living in dangerous areas. Instead they measure differences in mobility, social exposure, and embeddedness in conflictual interactions that the participation literature associates with higher political engagement (Gerber et al., 2011; Kam, 2012) and higher victimization risk (Cawvey et al., 2018). Some of the remaining differences, particularly in witnessing crime, may partly reflect micro-geographic sorting within census tracts rather than individual behavioral profiles. But the mobility and conflict results are difficult to explain solely through geography.

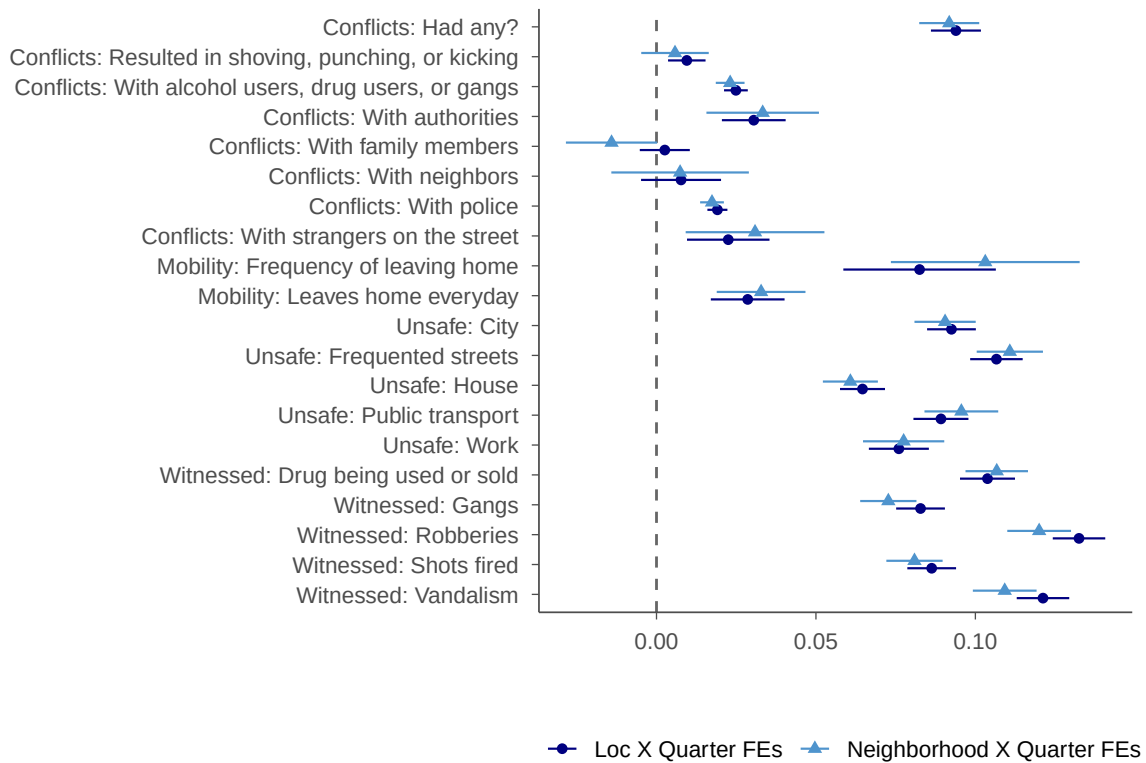


Figure 3: Cross-sectional differences between future victims and non-victims living in the same locality (dark blue) and the same neighborhood/census tract (light blue) on self-reported outcomes relating to exposure to crime. Robust standard errors clustered at the primary sampling unit. Full regression results in Table A14.

4.2 Reactions to Crime

I next estimate how victimization changes attitudes and behavior among those who experience it.

$$PM_{i[q]} = \beta_1 * Victim_{i[q]} + \theta_i + \gamma_q + \epsilon_{i[q]} \quad (4)$$

where $PM_{i[q]}$ is either a continuous measure of respondent i 's self-reported behavior, experiences, or attitude in year-quarter q , and neighborhood or locality g , or a binary measure that takes the value of 1 if respondent i reports that type of behavior, experience or attitude and 0 otherwise. $Victim_{i[q]}$ is an indicator variable that takes the value of 1 when respondent i reports having been victimized in the past six months in quarter q and 0 otherwise. Because victimization is asked every six months, while response variables are asked every three months, each respondent's victimization report is extended to cover the prior period. θ_i are individual fixed effects while γ_q are period fixed effects. $\epsilon_{i[q]}$ are robust errors clustered at the primary sampling unit. I use a TWFE estimator of the ATT as well as the counterfactual estimator with equal weights proposed by [Liu, Wang and Xu \(2022\)](#).

These estimators target the effect of victimization among those who experience it, an ATT for each behavior. Identification rests on parallel trends: had victimization not occurred, the outcomes of those who become victims would, on average, have changed as the outcomes of never-victimized respondents did, and victimization must not alter behavior before it occurs. In the framework's language, it suffices that response types be stable over the observation window.

Victimization changes attitudes and behaviors in a way consistent with demobilization, regardless of the preferred estimator (Figure 4). Compared with their behavior before becoming crime victims, respondents report leaving home less frequently, visiting friends and family less often, and taking more steps to prevent crime. They become less mobile and less likely to say they leave home every day. They trust law enforcement less, fear crime more, and adopt more constrained behaviors.

These behavioral and attitudinal changes are mechanisms through which demobilization is theorized to operate. Reduced mobility and social withdrawal limit the interpersonal interactions that facilitate political engagement ([Gerber et al., 2011](#); [Kam, 2012](#)). Increased fear raises the perceived costs of public engagement ([Ley, 2018](#); [Skogan, 1986](#)). ENSU does not measure political participation directly, but the within-individual

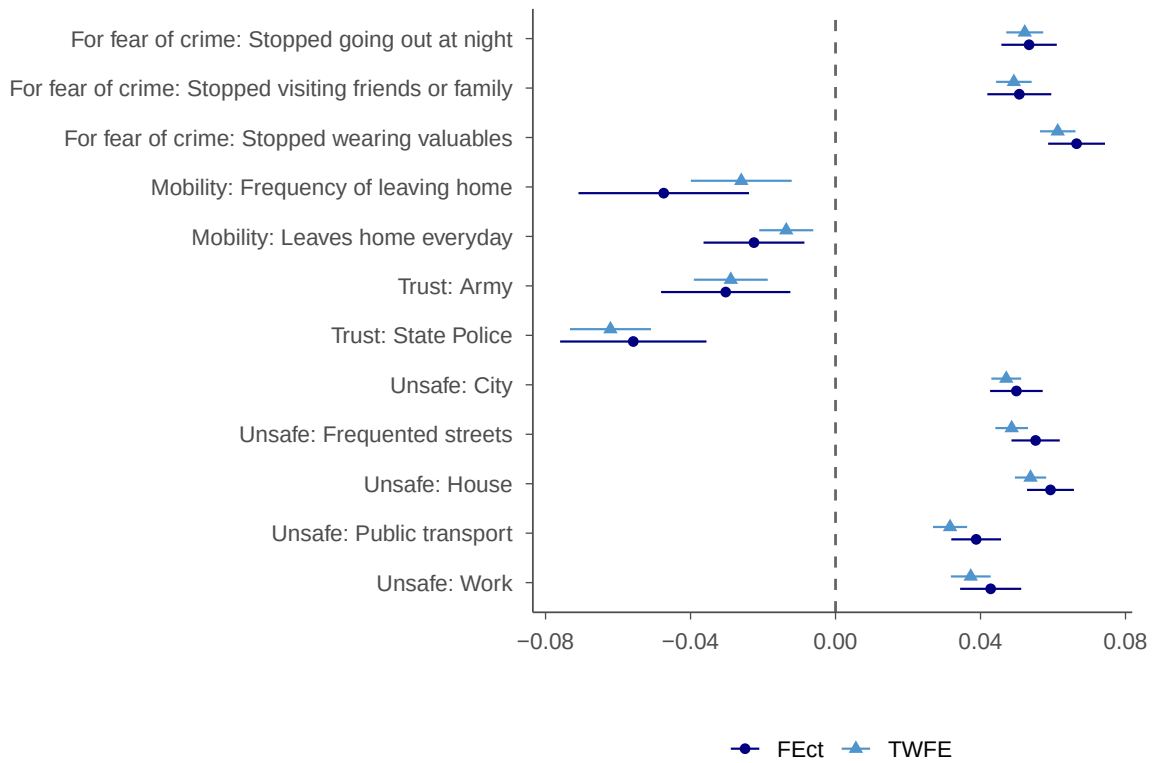


Figure 4: Within-individual effects of victimization on self-reported behavioral and attitudinal measures, estimated with the fixed-effect counterfactual estimator from [Liu, Wang and Xu \(2022\)](#) (dark blue) and a two-way fixed effects estimator (light blue). Robust standard errors clustered at the primary sampling unit. Full regression results in Table [A15](#).

changes it documents consistently move in the direction that the participation literature predicts should reduce engagement. The step from these behavioral changes to participation itself relies on the mapping that literature supplies, an assumption of the analysis rather than a result of it.

5 Disciplining the Identified Set with Panel Evidence

The LAPOP cross-section mapped the range of effects compatible with the data in a pooled sample from Latin American countries. The ENSU panel showed that, in Mexico, individuals with the behavioral profile of demobilizers are more exposed to crime, and that victimization moves behavior in the demobilizing direction. If this pattern generalizes across the region, the remaining question is whether mobilizers are numerous enough relative to demobilizers to keep the ATT positive despite their lower exposure. To address this question, recall from Section 1 that whether the ATT agrees in sign with the ATE depends on two quantities:

1. The *differential selection ratio*, $\lambda = \rho_U / \rho_T$: how much more likely demobilizers are to be victimized compared to mobilizers. When $\lambda = 1$, both strata face identical victimization probabilities and sign agreement is guaranteed (Corollary 2). When $\lambda > 1$, demobilizers are disproportionately exposed to treatment.
2. The *stratum-size ratio*, $\eta = \pi_T / \pi_U$: how numerous mobilizers are relative to demobilizers. When $\eta > 1$, mobilizers outnumber demobilizers in the population.

When $ATE > 0$, Proposition 3 establishes that the ATT is negative if and only if $\lambda > \eta$. That is, the ATT reverses sign relative to the ATE when the overexposure of demobilizers exceeds their relative scarcity. The ENSU panel showed that in Mexico, future victims are systematically more exposed than non-victims before victimization, and that victimization moves behavior in the demobilizing direction within individuals. Together, these findings are consistent with $\lambda > 1$ for Mexico: demobilizers likely face a higher risk of victimization than mobilizers.

I assume this directional condition holds across Latin America, without imposing any additional assumptions on the magnitude of λ . Given $\lambda > 1$, the sign of the ATT depends entirely on η , the ratio of mobilizers to demobilizers in the population. I therefore ask how restrictive the upper bound on η must be for *all* of the millions of ATTs for non-electoral participation outcomes compatible with the data to be neg-

ative. When 100% of compatible vectors agree on direction, the ATT is *sign-identified*. I restrict attention to LAPOP-compatible parameter vectors with $ATE > 0$ and $\lambda > 1$, gradually tighten the upper bound on η , and compute at each bound the proportion of vectors with $ATT < 0$. Table 5 shows the results.

	Bound on $\eta = \pi_T/\pi_U$ (mobilizers per demobilizer)				
	Unrestricted	≤ 2	≤ 1.5	≤ 1.1	≤ 1.04
<i>Share of feasible vectors with $ATT < 0$, given $ATE > 0$ and $\lambda > 1$</i>					
Protest	0.91	0.99	0.99	Signed –	Signed –
Community Problem-Solving	0.69	0.77	0.84	0.93	0.94
Community Meetings	0.80	0.90	0.94	0.98	0.99
Political Meetings	0.89	0.97	0.98	>0.99	Signed –

Table 5: Sensitivity of ATT sign identification to restrictions on the stratum-size ratio. Cells report the share of LAPOP-compatible parameter vectors with $ATT < 0$, conditional on $ATE > 0$ and $\lambda > 1$. Columns impose progressively tighter upper bounds on $\eta = \pi_T/\pi_U$.

Even without restricting η , imposing only $\lambda > 1$, between 69% and 91% of parameter vectors yield a negative ATT across the four non-electoral outcomes. At $\eta \leq 1.5$, all four outcomes exceed 84%. At $\eta \leq 1.1$, protest is signed negative and political meetings exceed 99%. At $\eta \leq 1.04$, protest and political meetings are signed negative, community meetings reach 99%, and community problem-solving 94%. These caps are restrictive relative to the unrestricted set, in which the median of η ranges from roughly 1.7 to 3.5 across outcomes (Figure 2). The sign conclusions therefore hold under the panel-motivated direction of selection and the stated bounds on η , not from the survey moments alone. Overall, results show that the mobilization finding for non-electoral participation is unlikely to hold up when shifting from the population average to those actually victimized.

As for the only electoral participation outcome, self-reported willingness to vote in the next election, the partial identification exercise and extant evidence suggest the ordering of treatment risks might reverse. First, the results from the partial identification procedure show the identified set concentrates on negative ATEs and positive ATTs. Indeed, Sønderskov et al. (2022) find consistent evidence: their within-individual estimate of victimization on turnout is positive, whereas the between-individual contrast in the same data is

negative, consistent with the T stratum being at higher risk of victimization. I therefore repeat the sensitivity exercise for that outcome, constraining the set instead to $\rho_T > \rho_U$ and asking how tight the bound on the number of demobilizers per mobilizer must be before the ATT on voting is signed positive given a negative ATE. I report the results in [A5.1](#). I find that the ATT of criminal victimization on willingness to vote is signed positive under mild bounds on that ratio.

6 Applications to other Forms of Violence

Criminal victimization is not the only form of violence where the risk of exposure may covary with potential response. I apply the partial identification procedure to reassess findings on the participatory consequences of civil war abduction in Uganda and police contact in the United States. Unlike the crime application, I cannot constrain the identified sets with panel evidence in these settings. The extensions, therefore, cannot establish sign reversal with the same force. What they can establish is structural consistency. In both settings, the region of the parameter space where the ATE and ATT disagree in sign is large, and the direction of disagreement mirrors the crime application.

6.1 Civil War Violence

Studies of civil war violence have generally found that exposure increases political participation, often attributing the increase to post-traumatic growth and increased prosociality ([Bellows and Miguel, 2009](#); [Blattman, 2009](#); [Voors et al., 2012](#)). Complementary work emphasizes that civil war violence can also reshape political identities, generating durable forms of political engagement rooted in conflict experience ([Balcells, 2012](#)).

One influential paper on the participatory consequences of civil war violence is [Blattman \(2009\)](#), which examines how forced recruitment into the LRA insurgency in Northern Uganda shaped downstream political participation. Leveraging quasi-random variation in abduction, the author targets the ATE of abduction on a battery of outcomes including participatory outcomes and finds that abduction significantly increased voting and doubled the likelihood of becoming a community leader.

I apply the same partial-identification procedure used for criminal victimization to ask whether the positive findings on the participatory effects of abduction also describe what happens to those abducted. Appendix section [A7.1](#) reports the full results. I find that the identified sets for both the ATE and the ATT

lean negative when the outcome is any form of non-electoral participation. However, the concentration of negatively signed estimands is much stronger for the ATT. Specifically, across peace-group membership, community mobilization, and volunteering, between 64% and 68% of compatible ATEs are negative, compared with 93% to 96% of compatible ATTs. These results contrast with the original findings, which report a positive and significant effect on community leadership and null effects for the other non-electoral outcomes. They also arise without imposing any restriction on response-type prevalence or abduction risk. As in the crime application, electoral participation shows a different sign pattern. For voting, both estimands point toward mobilization. 77% of compatible ATEs and 75% of compatible ATTs are positive.

Read against the original estimates, the positive community-leadership result reported in the paper is therefore compatible with a wide range of underlying causal structures, most of which imply a negative effect among those abducted. The null estimates for other forms of non-electoral activity are likewise compatible with strongly negative effects among the abducted. For voting, by contrast, the ATE and ATT mostly agree with each other and with the original mobilization finding.

The results should also be read in light of the quasi-random assignment claim in the original study. If abduction risk is unrelated to response type, as the author argues, the ATE and ATT coincide by Corollary 1, and the original estimates describe both the population and those abducted. The partial-identification exercise relaxes that restriction by allowing abduction risk to covary with potential response. It does not establish that abduction was selective or imply that the original design was flawed. It clarifies how much the substantive conclusion depends on the assumption that abduction was quasi-random in the presence of selection on responses. Last, these results exemplify that the partial identification procedure developed here does not mechanically generate either reversals or negative effects.

6.2 Contact with the Police

Research on the participatory consequences of police contact presents competing findings. Some work documents that contact with police teaches individuals they are politically marginal, suppressing engagement (Weaver and Lerman, 2010; Soss and Weaver, 2017). Others find that police contact, whether direct or indirect, stimulates participation (Walker, 2014, 2020). That risk of police contact is heterogeneous is a well-established fact. Police engage in statistical discrimination when stopping, and police contact is heterogeneously distributed along margins that covary with participation (Pierson et al., 2020; Ba et al., 2021;

Mummolo, 2017). Consequently, contact with police is potentially selective on participatory responses.

Walker (2020) finds that direct and indirect police contact is associated with higher rates of both electoral and non-electoral participation, findings read as mobilizing effects of contact for the sample at large and, more strongly, for nonwhites. I apply the partial identification procedure to one of its data sources, the National Crime and Politics Survey, a nationally representative U.S. survey conducted in 2013. I focus on direct police contact and use all available measures of electoral and non-electoral participation. Results are reported in Table A13.

The pattern here mirrors the crime application more closely. For most non-electoral outcomes, most parameter sets compatible with the data yield a positive ATE and a negative ATT. Fund-raising is the exception, with the sets dividing close to evenly and leaning the opposite way. For voting, the pattern reverses again: 89% of compatible parameter sets yield a negative ATE, while only 7% yield a negative ATT. If police contact were random, the sets imply it would increase non-electoral participation. If it is instead selectively experienced, in ways that coincide with heterogeneity in reactions, as the literature suggests, the identified set is consistent with contact depressing non-electoral engagement among those most frequently stopped while increasing their electoral participation. Under that reading, the substantive conclusion reverses. The contact that reads as mobilizing for the sample at large may demobilize, outside the voting booth, the communities most exposed to it.

7 Implications for Empirical Research

The framework and empirical results yield three implications for research on violence and other selectively borne harms. The first concerns which causal effect to target. The second explains when selection on responses, the dependence the methods literature calls essential heterogeneity or selection on gains (Heckman, Urzua and Vytlacil, 2006; Roy, 1951; Heckman, 1997), can make different estimands support different substantive conclusions. The third shows how the same dependence changes what researchers must assume to identify the effect of interest.

For selectively borne harms, the ATT should ordinarily be the benchmark estimand. An estimand specifies both the counterfactual contrast of interest and the population over which that contrast is averaged (Lundberg, Johnson and Stewart, 2021). It should be chosen to answer the substantive question, rather than inherited from the design or estimator. For selectively borne harms, the question of interest is usually what

the harm does to those who bear it, relative to a world in which they were spared it. The causal contrast corresponding to that question is the ATT, and research on the consequences of such harms should therefore treat the ATT as the benchmark estimand.

Other estimands answer other causal questions. The ATE, for example, describes the average effect, were exposure to be universal. That contrast may be the right target for applications interested in universal exposure counterfactuals. Researchers who target estimands other than the ATT where risk is unevenly borne, and where much of the population may never bear the harm, should say why their chosen counterfactual is theoretically relevant.

Importantly, narrowing the target population by selecting on observables does not render the distinction between estimands or counterfactuals moot. An ATE among young men, residents of high-risk neighborhoods, or any other observed subgroup still averages over exposed and unexposed units. If exposure remains selective across response types within that subgroup, its ATE and ATT may still differ.¹⁶ People who resemble victims are not people who were victimized.

Selection on responses makes estimand choice substantively consequential. Different estimands average individual treatment effects over different populations. They therefore answer different causal questions and need not agree (Heckman and Vytlačil, 2005; Lundberg, Johnson and Stewart, 2021; Egami and Hartman, 2023). The distinction becomes especially consequential when individual effects can run in opposite directions and treatment risk differs across response types. Alternative estimands may then differ not only in magnitude but also in sign.

For the ATE and ATT, the framework shows how selective treatment risk changes the balance of mobilizing and demobilizing responses being averaged. The ATE depends on the relative prevalence of positive and negative response types in the population. The ATT depends on those same population shares, weighted by the treatment risk of each type. The two estimands therefore aggregate the same response structure under different weights. Estimators left to their defaults may apply weights that correspond to neither target population (Aronow and Samii, 2016; Słoczyński, 2022). When treatment falls disproportionately on one response type, the balance of positive and negative effects among the treated can differ from the balance in

¹⁶Within any subgroup S , $ATE_S = \pi_T^S - \pi_U^S$, while the ATT_S reweights those shares by within- S exposure probabilities. Corollaries 1 and 2 apply within S as in the population at large: the two contrasts coincide only where exposure is unselective across response types within S .

the population. That dependence between treatment risk and potential response is selection on responses.

The conditions that produce this divergence may arise across many settings in empirical political science. Many treatments central to the discipline are theorized to provoke both engagement and withdrawal, support and backlash, compliance and resistance. Treatment risk is also shaped by institutions, geography, resources, behavior, and political opportunity, many of the same forces that shape how people respond. Researchers should therefore ask whether those at greatest risk of treatment may respond differently from those at lower risk. When that possibility is credible, and the substantive question concerns those exposed, they should prioritize the ATT. At minimum, researchers should assess whether selection on responses could reverse the sign of the effect and restrict directional claims to the estimand actually recovered.

Selection on responses also complicates identification. The central argument of the paper concerns estimand choice rather than identification. For selectively borne harms, the population-wide effect often targets a counterfactual of universal exposure that is unlikely to arise and of limited theoretical interest. When treatment risk also covaries with potential response, that effect may differ in sign from the effect among those harmed. The empirical analyses thus far have therefore bracketed identification and asked what follows from changing the target estimand.

However, the same selection on responses that can produce sign reversals also makes common identifying assumptions more demanding. In general, identification must account for selection in both untreated outcomes and treatment responses (Xie, 2013; Heckman and Vytlačil, 2005). That is, for the possibility that treated and untreated units differ in what they would have done without treatment *and* in how treatment would change their outcomes. Appendix A3.1 elaborates on these requirements.

Cross-sectional designs must address both forms of selection using observed covariates. For the conditional-ignorability designs that dominate the literature, identification requires treatment risk to be unrelated to the full pair of potential outcomes within covariate strata. In the language of the framework, mobilizers, demobilizers, always-participants, and never-participants must face the same treatment risk after conditioning. An adjustment set must therefore account not only for differences in untreated outcomes, but also for the relationship between treatment risk and the response treatment would produce. Whether observed covariates are sufficient is a stringent substantive claim that researchers should defend directly.

Panel designs replace cross-sectional exchangeability with assumptions about counterfactual change.

Such difference-in-differences designs require that treatment not alter outcomes before it occurs and that, absent treatment, the average outcome of those who become treated would have changed as the average outcome of the comparison group did. Substantively, for identification in such settings, selection on responses need not be ruled out. Treated units may differ in how treatment would affect them. Identification instead requires that the forces sorting them into treatment would not also have produced different untreated trajectories. When the identifying assumptions required by a design cannot be defended, researchers need not force point identification. They can instead characterize the effects consistent with weaker assumptions, as the partial-identification procedure developed here does, and state what additional evidence would narrow the identified set.

8 Conclusion

What are the participatory consequences of violence? Any answer to that question always requires a corollary: for whom? Yet empirical research on violence often treats the population over which its effects are defined as a secondary concern. I have argued instead that the choice of estimand is of first-order importance when investigating the effects of violence and other selectively borne harms. The substantive importance of estimand choice stems from the essentially disjoint potential consequences of violence. Violence can spur some into political action and demobilize others. If victimization risk covaries with such responses to violence, then what violence does to victims need not be what it would do to those it spares.

More concretely, I argue that existing findings suggesting crime mobilizes may belie demobilization among victims. Indeed, the empirical results here presented do not support the claim that crime mobilizes its victims. The partial-identification exercise shows which combinations of the ATE and the ATT are jointly compatible with the observed rates of victimization and participation in Latin America. For non-electoral participation, most feasible pairs combine a positive population-wide effect with a negative effect among victims. The Mexican panel provides evidence of a selection pattern in which demobilizers are at higher risk of crime, consistent with the observed estimand divergence resulting from the partial identification exercise. Before victimization, future victims were more mobile, more exposed to crime, and more involved in conflict than their neighbors. After victimization, they became less mobile, more fearful, and less trusting of law enforcement. Taken together, these results suggest that people at greater risk of becoming a crime victim are also more likely to withdraw after it occurs. Population-wide estimates can therefore indicate

mobilization even when victimization reduces participation among victims.

A natural question is why people who withdraw after victimization would face greater exposure to crime in the first place. Unlike many settings with selection on treatment effects, victims do not choose crime. Yet, selection on responses need not require individual choice to materialize in the political sphere. Crime risk is shaped by where people live, the resources they possess, the institutions that protect them, and the routines and constraints that govern their daily lives. These same conditions also shape the political responses available to them after victimization. They influence whether people can seek protection, make claims on the state, remain active in public life, or instead retreat from it. Treatment risk can therefore covary with potential response, even when no one deliberately selects into treatment.

The finding that crime can appear politically mobilizing in the population even as it demobilizes the people who are actually victimized has important substantive implications for the study of the political economy of crime. When those at greatest risk are also the most likely to withdraw from political life after victimization, governments face weaker incentives to protect the people who bear the greatest burden of violence. Security policy may instead respond disproportionately to citizens who are less exposed, better protected, and better able to turn experiences of crime into political claims. In Latin America, one of the world's most violent and unequal regions and the setting of these findings ([Vilalta, 2020](#); [Gasparini and Lustig, 2011](#)), crime may deepen the same inequalities that structure differential exposure to violence.

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